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Text Analytics

From Detection to Diagnosis: On the Conditional Suitability of Large Language Models for Intrinsic Hallucination Detection

A Seminar Paper on Capabilities, Limits, and Future Directions

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March 7, 2026

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1 Introduction

With the rapid advancement in Large Language Models (LLMs) the human-computer interaction is fundamentally transformed. Models like ChatGPT, Llama, and Claude are capable of producing text that is grammatically fluent, stylistically coherent, and basically indistinguishable from human writing. However, this surface-level fluency hides a fundamental flaw. LLMs frequently generate content that on first sight appears plausible due to the convincing nature but is either factually incorrect or unsupported by the provided source material. The literature refers to this phenomenon as *hallucination*. As AI and LLMs are more and more incorporated in everyday life raising concerns come up about the trustworthiness of these new technologies. Especially the deployment in high stakes domains such as the medical sector, legal consultation or in educational support reliability and safety are critical. The consequences of such hallucinations can range from minor inaccuracies to serious misinformation impacting human lives. What makes this problem particularly difficult to address is that hallucinated content is not always obviously wrong. A model may confidently cite a non-existing study, subtly misattribute a quote or generate a translation unfaithful to the source. The plausibility of hallucinated content is precisely what makes it dangerous. Users who lack domain expertise or access to the source material have no reliable means of detecting such errors on their own. Motivated by these risks, recent work emphasizes the need for automated detection and evaluation methods, especially in scenarios where human verification is infeasible (e.g., cross-lingual use cases where users cannot reliably assess translation correctness).

1.1 Seed Paper and Thesis

Building upon the challenges outlined above, this paper is centered around the seed paper *"Can LLMs Detect Intrinsic Hallucination in Paraphrasing and Machine Translation"* (Gogoulou et al. 2025), which investigates the inherent ability of LLMs to detect intrinsic hallucination in two scenarios: paraphrasing and machine translation. The paper frames this detection task as an entailment-based reasoning problem and systematically compares LLMs against Natural Language Inference (NLI) models, examining how factors such as model size, instruction-tuning, prompt language, and prompt formulation influence detection performance.

The central thesis of this paper will argue *that LLMs are conditionally capable of detecting intrinsic hallucination, but this capability is structurally insufficient for real world deployment*, motivating the discussion to shift from detection to diagnosis. To address this statement, the paper first establishes the theoretical groundwork by defining all necessary taxonomy Section 2. Section 4 then reviews alternative detection approaches and discusses their relevance for future hallucination detection systems. Building on these findings, Section 5 critically examines whether LLMs are inherently capable of reliably detecting intrinsic hallucinations, before the paper concludes by outlining which approaches offer the most promise for future development.

2 Theoretical Background and Key Concepts

2.1 Hallucination

In the real world, hallucination is defined as *perceptions in the absence of an external stimulus and are accompanied by a compelling sense of their reality.* (Boksa 2009, p.260). In the context of Language generation, hallucination refers to a model's output that on first sight seems fluent and plausible yet is either factually incorrect or not supported by a given source or context. In general does the specific definition of hallucination vary between studies yet this a common characterization.(Ji et al. 2023, p.248:3) defines hallucination as *the generated content that is nonsensical or unfaithful to the provided source content* which is widely adopted in the research community. It is therefore important to note that the concrete boundaries of what counts as hallucination are not universally defined. Different research communities and tasks apply the term with different weight and strictness. Hallucination substantially depends on the application context, a point which will be mentioned and discussed in greater detail in the following sections.

2.2 Intrinsic vs. Extrinsic Hallucination

In the context of LLMs, hallucination is commonly further divided into intrinsic and extrinsic hallucination.

Intrinsic hallucination generally describes cases where the output is deficient with respect to a particular input and or contradicts it. Crucially, can this deficiency be detected solely by comparing the input and output, without requiring external world knowledge (Ji et al. 2023, p.248:3). (Ji et al. 2023, p.248:3) illustrates this clearly: given the source sentence *"The first vaccine for Ebola was approved by the FDA in 2019"*, a model generating the summary *"The first Ebola vaccine was approved in 2021"* produces an intrinsic hallucination, as the date directly contradicts the source. Similarly, in a machine translation context, translating the German sentence *"Michael ging am Donnerstag in den Buchladen"* as *"Jerry didn't go to the bookstore"* leads to an intrinsic hallucination, since both the subject and the negation contradict the original input. The inconsistency is fully self-contained within the input-output pair and can in principle be detected through semantic consistency reasoning alone. It is important to mention that a faithful output can be factually incorrect.

Extrinsic hallucination on the other hand, describes cases that involve information that is not present in the input and cannot be verified solely based on the source alone. In order to detect extrinsic hallucination it typically requires an external knowledge source. Crucially it does not matter whether the part which is not entailed happens to be factually correct. Extrinsic hallucination can therefore still be accurate compared to world knowledge.

An example from (Ji et al. 2023, p.248:3) shows the concrete distinction between intrinsic and extrinsic hallucination. Looking at the same source text about the Ebola vaccine, a model generating the summary *"China has already started clinical trials of the COVID-19 vaccine"* will produce an extrinsic hallucination because neither the source nor any internal reasoning confirms or denies this claim. Likewise, for the machine translation scenario, translating *"Michael ging am Donnerstag in den Buchladen"* as *"Michael happily went to the bookstore with his friend"* constitutes an extrinsic hallucination. Adding the extra details

"happily" and *"with his friend"* introduce new information which the source cannot verify, even though the translation is not directly contradictory.

2.3 Faithfulness and Factuality

Terms also often used in the context of hallucination are factuality and faithfulness. Faithfulness describes how well a model's generated output is consistent by the given input or user provided context. Therefore faithfulness is closely related to intrinsic hallucination. (Ji et al. 2023, p.248:5) describes it as *staying consistent and truthful to the provided source - antonym to "hallucination"*. Thereby if we try to maximize faithfulness of an output it should contain no intrinsic hallucination. As intrinsic hallucination does not require any outside knowledge, faithfulness-oriented evaluation methods or other interference models are particularly well suited for this detection task. Especially in conditional generation tasks such as summarization, paraphrasing or machine translation.
On the contrary *Factuality* refers to the quality of being actual or based on fact. However, does the interpretation of the "fact" vary. Papers like (Dong et al. 2020) treat the source itself as the factual reference, effectively giving the term "factuality" and "faithfulness" the same meaning. For both the seed paper and this thesis, we adopt the definition proposed by (Maynez et al. 2020, p.3), which clearly distinguishes source knowledge from world knowledge.

2.4 Detection vs. Mitigation: Why Detection Matters

An increasing part of recent research focuses on hallucination mitigation, meaning that at the time of generation the output is being checked for hallucination. However, in many practical cases this is not enough, due to inherent problems such as biases, knowledge gaps, and architecture which are fundamentally embedded in the model (Ji et al. 2023, p. 248:31). Hallucination detection is therefore not merely a complement to mitigation but an independent research problem in its own right. Mitigation on one hand can leverage its internal states or generation process to intervene during generation. Detection on the other hand must solely reason post-hoc from the final input-output pair without access to the model that generated the text. Especially for the detection of hallucination, this means the detector must perform stable semantic consistency reasoning under no additional constraints. The following sections will show why detection poses a non-trivial problem.

2.5 The Complexity of Intrinsic Hallucination Detection

While intrinsic hallucination may appear to be a straightforward consistency check between source input and model output, both empirical evidence and theoretical analysis reveal that it constitutes a fundamentally non-trivial problem. On the empirical side, even for humans the JUDGE-BENCH (Bavaresco et al. 2025, p.241) study shows varying agreement not only between humans and models, but also among human annotators themself. The reports in Table 3 (pp.11-12), show that inter-annotator agreement (e.g. Krippendorff's alpha between

humans vary strongly for certain tasks, indicating that even humans frequently disagree on properties like coherence, consistency and faithfulness. These results suggest that intrinsic hallucination may not be a binary or objectively trivial problem but often requires interpretation. Especially in cases of faithfulness and entailment it is not trivial to determine whether a text should be considered to contain hallucination or not.

Furthermore, does (Ji et al. 2023, p. 248:4) note that the tolerance for what constitutes hallucination varies across tasks and domains: what is considered a faithful output in an open-domain dialogue system may be classified as intrinsic hallucination in a summarization or machine translation context. This definitional instability further adds to the difficulty of detection, as evaluation methods must be calibrated not only to a model's output but also to the circumstances of the specific task. Taken together, these considerations establish that intrinsic hallucination detection is neither a solved problem nor a simple classification task, but a challenge that operates simultaneously at the level of linguistic consistency, human judgment, and mathematical computability. These limitations motivate a more detailed investigation and will be discussed in the subsequent sections of this paper. The following section focuses on intrinsic hallucination detection and examines how this problem is addressed in the seed paper *Can LLMs Detect Intrinsic Hallucinations in Paraphrasing and Machine Translation?*.

3 Seed Paper at the Core: Detecting Intrinsic Hallucinations

This section presents the seed paper "Can LLMs Detect Intrinsic Hallucination in Paraphrasing and Machine Translation?" as the core of this seminar paper. The paper focuses on intrinsic hallucination and explores whether LLMs can reliably detect such errors.

3.1 Methods of paper

The seed paper formulates intrinsic hallucination as an entailment problem. An output is considered faithful if it is fully entailed by the source input, and it is considered to contain a hallucination if it introduces additional or contradictory information not logically supported by the source. This formulation allows the detection task to be treated as a reasoning task rather than a knowledge verification problem, since no external world knowledge is required and only the relationship between source and hypothesis needs to be accessed. To conduct this approach, the authors construct a contrastive evaluation setup. Each instance consists of a source sentence and two candidate hypotheses, where exactly one is faithful (hyp+) and the other contains intrinsic hallucination (hyp-). The model is then asked to determine which of the hypotheses is the hallucinated one. This setup ensures a controlled environment and allows for direct comparison across models. The methodology is grounded in the HalluciGen dataset (Dürlich et al. 2024), which was specifically designed for intrinsic hallucination detection and forms the empirical basis of this study. The overall evaluation pipeline is illustrated in Figure 1. As shown there, the source sentence together with both hypotheses are provided as context to either an LLM or an NLI model.

The dataset covers two scenarios: paraphrase generation (PG) and machine translation (MT). In the paraphrasing task, both source and hypotheses are in the same language. Either in English or Swedish. The translation task is cross-lingual covering English-French and English-German in both translation directions. Additionally is every hallucinated hypothesis in the dataset assigned to one of eleven categories that describe the type of error breaking the entailment relation: antonym, tense, negation, etc. (Gogoulou et al. 2025, p.11)).

To evaluate how different models respond to varying prompt designs, the study tests six prompt formulations. Some prompts explicitly reference the concept of hallucinations and, in some cases, also provide a definition, while others avoid the term and instead describe the task indirectly. For example, Prompt 4 is formulated as: *"Given a source sentence (src) and two <scenario> hypotheses (hyp1 and hyp2), detect which one of the two logically contradicts the src."* (Gogoulou et al. 2025, p.5). Lastly to optimize outputs and ensure proper formatting in addition to the prompts every model receives the same instructions. Models are instructed to only answer with either "hyp1" or "hyp2". Responses claiming that both or neither hypothesis are hallucinated, or answers that contained an explanation were treated as invalid and excluded from the evaluation. The Study chose ten open-access LLMs across four different model families: Llama 3, Mixtral, EuroLLM, and GPT-SW3 also varying in model size and the degree of instruction tuning (Gogoulou et al. 2025, Table 2, p.8). This specific selection enabled a systematic comparison of how model size, instruction tuning and prompt formulation impacts the detection performance. The authors used NLI models as the baselines to directly compare their detection capabilities. Natural Language Inference (NLI) models are classifiers trained to determine the logical relationship between a premise and a hypothesis, typically assigning one of three labels: entailment, contradiction or neutral. This makes them a natural fit for intrinsic hallucination, where the core question is whether a hypothesis is faithful to a given source. Previous work like, (Manakul, Liusie, and Gales 2023) demonstrate in their SelfCheckGPT framework that NLI-based hallucination detection performs comparably to full LLM prompting, with only a marginal gap between the two approaches. Considering their computational efficiency and task alignment, NLI models serve as a strong baseline for this experiment. Lastly were all models evaluated using the F1 score against gold labels from HalluciGen dataset.

3.2 Results of the Paper

The results of (Gogoulou et al. 2025, p.4-8) yield several findings central to the thesis of this paper. Among the evaluated LLMs, META-LLAMA-3-70B-INSTRUCT and MIXTRAL-8X7B-INSTRUCT emerge as the strongest competitors, achieving the highest F1 scores in both scenarios. The performance of smaller models like EUROLLM-1.7B and GPT-SW3 is significantly lower and in some cases drops to random baseline levels. In the translation scenario, the strongest LLMs tend to outperform the NLI baseline in most language directions. However in the paraphrasing scenario the NLI baseline BGE-M3-ZEROSHOT-V2.0 achieves a F1 score of around 0.90, outperforming all evaluated LLMs. Looking at the four investigated factors — model size, instruction tuning, prompt language and prompt formulation — none of them can serve as a reliable predictor. Each factor affects the performance differently across all model families. Regarding model size, no certain trend is detectable. Within

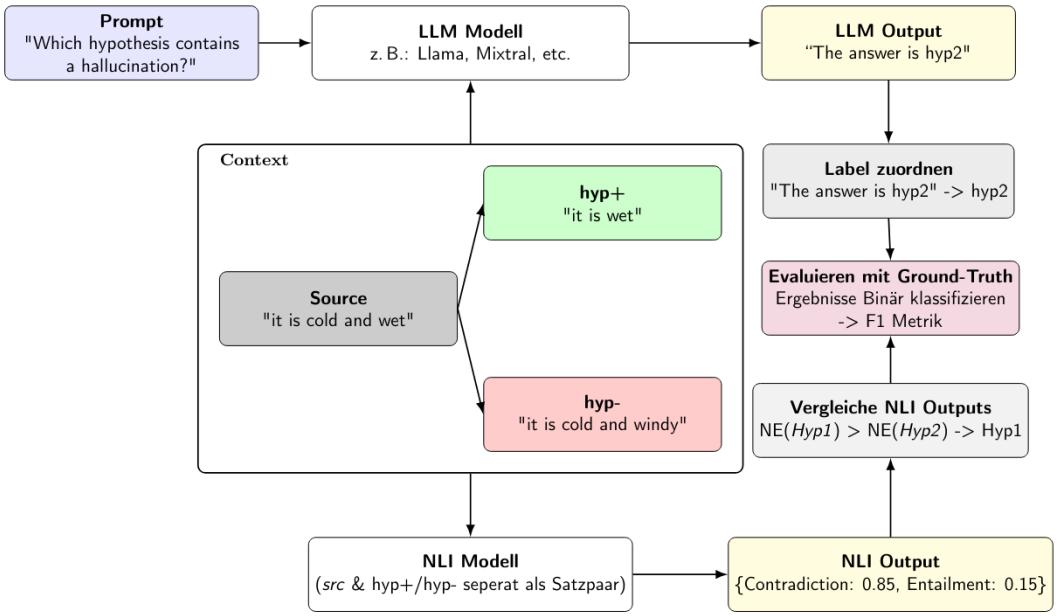

Figure 1: Conceptual pipeline for hallucination detection following the setup proposed by (Gogoulou et al. (2025) p.2-5). Given a source sentence and two hypotheses (hyp+, hyp-), an LLM is prompted to identify which hypothesis contains a hallucination. The model output is mapped to a label and evaluated against the ground-truth using an F1-based classification metric. In addition, an NLI model can be used as a baseline by comparing entailment scores between the source and each hypothesis.

the Llama3 family, the larger 70B variant clearly outperforms the 8B model. On the other side in the Mixtral family, scaling the 8x7B to the 8x20B model consistently worsens performance across all tested scenarios and languages, suggesting that bigger parameter size does not translate to better entailment reasoning. Looking at the impact of instruction tuning, reveals that a clear positive effect is only observed for the largest Llama3 model. For the other models like GPT-SW3, the instruction-tuned variant is more consistent than its base model. Regarding prompt formulation and language, performance remains decently stable across most models and prompt variations. However critical exceptions exist. Results from the model MIXTRAL-8X22B-INSTRUCT show a standard deviation of up to 0.23 across all prompt variations, rendering the model unreliable for this task. Furthermore, including the explicit term *hallucination* in the prompt tends to negatively impact the models performance. Considering all the inconsistencies across all four factors suggest that none serve as a reliable indicator for better results. Performance appears to emerge from a complex and model-specific interaction of all mentioned variables.

All these findings conclude to a picture that is central to the broader argument of this thesis. On one hand, do the results confirm that LLMs possess the inherent ability to detect intrinsic hallucination. Particularly in cross-lingual settings where the multilingual pretraining provides a structural advantage over discriminative NLI models. On the other hand, did the results show that performance varies strongly across task, model and prompt formu-

lation.

It is however important to note, that these results must be interpreted with caution, as all results were obtained under a highly controlled contrastive detection scenario. Each evaluation instance always presented the model with exactly one faithful and one hallucinated hypothesis, reducing the detection problem to a binary choice. In theory this setup is methodologically clean and structurally executed, but it shifts significantly from real-world detection conditions, where mostly no contrastive pair is available and the model must access from a single output. While the results confirm that LLMs are inherently capable of detecting intrinsic hallucination, this capability has to be put into perspective. The controlled contrastive setup systematically inflates performance estimates by providing the model with a comparison that would not be available under real-world detection conditions. It is therefore important to mention that being capable of detecting hallucination under idealized conditions and being a reliable detector in deployment are not the same thing.

These results have an implication for the thesis of this paper. The conditional suitability of LLMs demonstrated by (Gogoulou et al. 2025) should be understood as an upper bound on performance under idealized conditions, not as proof of general-purpose detection capability. The following sections will examine whether this limitation is structural, and what alternative or complementary approaches may offer a more robust foundation and how LLMs can be used differently for the detection of intrinsic hallucination.

4 Alternative Approaches to Intrinsic Hallucination Detection

The general results of the seed paper established that LLMs are conditionally capable of detecting intrinsic hallucination, but fail to guarantee reliability. This raises the natural question: *what can alternative detection approaches contribute which LLMs alone can not consistently provide?* To answer this, the following section reviews the most relevant alternative approaches beyond prompt-based LLM evaluation, assessing each method with respect to their specific strengths and the gaps they leave open. The goal is to highlight why no single method is sufficient on its own, and why complementary weaknesses of all approaches directly motivate hybrid solutions.

4.1 Entailment-Based Approaches

As briefly established in the seed paper, intrinsic hallucination detection is generally understood as a semantic consistency problem meaning a faithful output must be entailed by its source. NLI models are the most direct instantiation of this idea, and early work such as (Maynez et al. 2020, p. 1910) demonstrated that pretrained NLI models can reliably identify faithfulness violations in abstractive summarization. Their main strength lies in their specific training to reason about entailment or contradiction. This ability lends itself well for the detection of intrinsic hallucination as entailment and faithfulness are closely related concepts. However, as the results of (Gogoulou et al. 2025, p.5) revealed what NLI baseline performance is strong in monolingual paraphrase, its performance degrades in cross-lingual translation settings. Moreover does the NLI-based detection rely on the existence of a clear, explicit source-output pair. This condition may hold for controlled scenarios like paraphrasing or

machine translation, but for open-ended settings such as dialogue or question answering, where no single faithful reference may exist, this approach could collapse. This suggests that a purely entailment-based approach requires augmentation through additional detection signals to generalize beyond controlled sentence-level scenarios.

4.2 Self-Consistency and Uncertainty-Based Methods

Another method to detect intrinsic hallucination is self-consistency-based-detection. One promising system is (Manakul, Liusie, and Gales 2023) SelfCheckGPT, a framework combining LLM prompting, token-level probability signals and NLI-based entailment checks to detect hallucination on a sentence-level. The main idea of this approach is to generate multiple responses for the same input and evaluate the internal agreement. If the outputs of the model contradict each other or vary at certain parts it could imply hallucination. Although the paper does not explicitly differentiate between intrinsic and extrinsic hallucination, its reliance on entailment checks between source and output and the avoidance of external knowledge comparisons makes it particularly applicable for intrinsic inconsistencies. This work is closely related to uncertainty-based and LLM-based self-evaluation approaches. These methods assume that hallucinated content occurs from higher predictive uncertainty. These uncertainties can be then measured through entropy, token-level log probabilities or variance across multiple generations. Nevertheless this approach has its flaws as it deduces that high confidence correlates with faithfulness. This may lead to cases where small logical errors like negation shifts, incorrect entities or numerical errors stay undetected due to the high confidence of the model.

4.3 Embedding-Based Detection

Embedding-based methods measure semantic similarity between source and output in a continuous vector space, with the assumption that faithful outputs remain semantically close to their source. Similarity-based approaches such as those using cross-lingual models like LaBSE have shown promise for machine translation hallucination detection specifically. (Hu et al. 2024) further combine embedding shifts with uncertainty signals to improve detection robustness. The strength of this paradigm lies in its language-agnostic nature and computational efficiency, making it well suited for cross-lingual settings like those evaluated in the seed paper. Its limitation, however, is that coarse semantic similarity can miss locally confined but semantically significant errors. A small negation shift or a named entity substitution may leave the overall embedding largely unchanged while fundamentally altering the meaning.

4.4 Comparative Summary

Taken all these approaches together, they all capture different dimensions of semantic inconsistency and exhibit complementary strengths and weaknesses. NLI models offer principled, stable entailment reasoning but are limited in granularity and fail to scale over all tasks. Uncertainty-based methods provide an approach that does not require training but fails to signal hallucination on high-confidence errors. Embedding-based methods scale well

across languages but miss locally confined inconsistencies. Prompt-based LLM detection offers flexibility and strong out-of-the-box performance but lacks reliability and stability. Most approaches suffer from the lack of semantic reasoning creating their decision solely on probabilities which will miss subtle changes. Crucially, no single method dominates across all hallucination types, tasks and languages, a finding that directly motivates the hybrid architectures discussed in Section 6, where complementary signals are combined to compensate for the individual limitations of each approach.

5 Discussion: The Limits of LLM-Based Detection?

5.1 LLM-as-Judge Perspective

Building upon the idea of using LLMs as a detection method for intrinsic hallucination, a more fundamental question arises: *how well are LLMs inherently suited to evaluate faithfulness?* In order to properly address this, it is useful to situate the detection task within the broader paradigm of LLM-as-a-Judge. At its basic core, does LLM-as-a-Judge refer to the use of large language models as automated evaluators to assess generated text for criteria such as quality, correctness, or faithfulness. Before the era of LLMs, evaluation of natural language generation was driven by either costly and slow human evaluation, or by automatic metrics such as BLEU or ROUGE, which provided consistency yet failed to capture deeper semantic relationships. The increasing ability of LLMs to mimic human reasoning now enables automated evaluation across a wide range of tasks and domains without the bottleneck of human annotation (Gu et al. 2025, p.2). Crucially, research consistently confirms that LLM-based evaluation is particularly well suited for faithfulness assessment. (Malin et al. 2024, p.11) find across summarization, question-answering, and machine translation tasks that LLM evaluators most consistently achieve the highest correlation with human judgement, outperforming n-gram, embedding-based, and fact-extraction approaches. This advantage stems from the LLM's ability to capture nuanced semantic relationships and entailment patterns that surface-level metrics systematically miss. This finding is further corroborated by (Yang Liu et al. 2023, p.1), whose G-Eval framework demonstrates that GPT-4 achieves the highest Spearman correlation with human judgements on faithfulness-related criteria, surpassing all previously established automatic metrics. Instruction-tuned LLMs have been exposed to vast quantities of text implicitly encoding entailment relationships and consistency norms, enabling them to internalize a form of semantic judgment that closely approximates human reasoning. (Gogoulou et al. 2025) further confirm this, showing that a well-designed prompt alone is sufficient to guide a model toward structured faithfulness judgments in a contrastive detection setting, particularly in cross-lingual scenarios where multilingual pretraining provides a structural advantage over discriminative NLI models. Taken together, these findings establish that LLMs are not merely a convenient substitute for human evaluators but represent the current state of the art in automated faithfulness evaluation.

Despite this strong evidence in favour of LLMs as faithfulness evaluators, their suitability cannot be considered unconditional. The same architectural properties that allows LLMs to approximate human semantic reasoning also introduce systematic vulnerabilities which directly affect the reliability of their judgments. These weaknesses are not incidental im-

plementation flaws but are rooted in how LLMs are trained and how they generate output, making them relevant regardless of the specific evaluation task. The following subsection examines these systematic biases and their concrete implications for intrinsic hallucination detection.

5.2 Systematic biases

Another source of unreliability comes from systematic biases introduced in the training process. (Alansari and Luqman 2025, p.8) identify two particularly relevant failure modes at the fine-tuning stage: capability misalignment, where models are trained to give confident answers even when their knowledge is insufficient, and sycophantic behavior, where models generate outputs that appear plausible and evaluator-pleasing rather than faithfully grounded. Both tendencies directly undermine the reliability of LLMs as hallucination detectors. A model trained to appear confident and coherent will systematically underreport uncertainty and may favor plausible-sounding outputs over faithful ones. This concern is reinforced by (Bavaresco et al. 2025 p. 241), who find that LLM judges correlate more strongly with non-expert human annotations than with expert judgments, suggesting that LLMs may be approximating surface believability rather than performing genuine faithfulness evaluation.

Prompt sensitivity as an instability indicator. The seedpaper (Gogoulou et al. 2025) provides direct empirical evidence of a further weakness: the sensitivity of LLM performance to prompt formulation. MIXTRAL-8X22B-INSTRUCT shows a standard deviation of up to 0.23 across prompt variations in the Swedish paraphrase scenario, a range so large it renders the model unreliable for this task (Gogoulou et al. 2025, p.6, Table 2). Notably, including the term "hallucination" in the prompt negatively impacts performance for several models, suggesting that the models' internal representations of the concept do not align consistently with its operational definition in the detection task. This prompt sensitivity is not a minor engineering concern but reveals that LLM-based detection is not a stable and principled evaluation procedure but one whose results depend heavily on framing choices external to the model itself. (Alansari and Luqman 2025, p. 1) arrive at a structurally similar conclusion from a broader perspective, noting that no single detection approach performs well under all circumstances and that the inconsistency in prompting techniques is itself a barrier to developing universal detection frameworks.

5.3 Theoretical Limitations

While the systematic biases discussed above arise primarily from training-time decisions and can in principle be addressed through improved fine-tuning or prompt engineering, a second class of limitations runs deeper. These are not artifacts of how a particular model was trained, but consequences of the fundamental mathematical structure underlying all autoregressive language models. Even a perfectly fine-tuned LLM, free of sycophantic tendencies and robust to prompt variation, would remain subject to the following theoretical constraints.

The recursive problem of self-evaluation. The most fundamental limitation with using LLMs as the hallucination detector is them being prone to the same biases as the evaluated

text itself. LLMs themself are generative probabilistic models and therefore susceptible to producing hallucinated content. (Banerjee, Agarwal, and Singla 2024, p.13) argues that hallucination is not an occasional phenomena but an error which stems from the mathematical and logical structure of large language models in general. Since LLMs work as an autoregressive system, predicting the next token based on learned probability distributions and previous context, rather than learned truths, they inherently do not possess the ability to guarantee correctness. Henceforth if such a system is tasked to detect hallucination in an output, which is generated by itself or another model, a recursive problem emerges. (Banerjee, Agarwal, and Singla 2024, p.20) goes further and demonstrates that hallucination is a structural problem, which exists at every stage of the LLM pipeline. Incomplete training data, undecidable information retrieval, unresolvable intent classification and the core unpredictability of token generation each contribute a non-zero probability of hallucination. Considering that all these stages in the pipeline are sequential only compounds the probabilities for failure. These proofs underline that even architectural improvements, larger datasets or post-hoc detection can only reduce but never fully eliminate hallucination. Even though (Banerjee, Agarwal, and Singla 2024, p.8) primarily describe factual inconsistencies and therefore extrinsic hallucination the arguments still apply for intrinsic hallucinations, albeit in a slightly different form. As intrinsic hallucination evaluates the faithfulness of an input in comparison to an output, detecting such inconsistencies still requires reliable intent interpretation, accurate retrieval of relevant source segments and first and foremost stable semantic comparisons between both texts. All of these properties are subject to uncertainty and if they cannot be entirely guaranteed due to the probabilistic nature of LLMs, evaluations can never be fully faithful. Even if the task is limited to only internal consistencies, the evaluation process itself remains grounded in probabilistic modelling and is therefore as shown in (Banerjee, Agarwal, and Singla 2024, p.20) mathematically unable to ensure perfect reliability.

5.4 Conclusion LLMs for detection

Taken together, the findings of the survey suggest a nuanced conclusion. LLMs are potentially suitable for hallucination detection because they combine semantic reasoning with scalability, outperforming shallow automated metrics . However, their suitability is conditional rather than absolute. Reliability depends on validated agreement with human experts, systematic bias mitigation, adversarial robustness testing, careful prompt engineering and possibly ensemble integration. Moreover, because LLM judges can themselves hallucinate, they cannot guarantee objective truth verification.

Therefore, LLMs should not be viewed as autonomous truth detectors but as probabilistic evaluators whose effectiveness in hallucination detection depends on rigorous reliability engineering. They are suitable under controlled, validated conditions, but unsuitable as standalone, uncalibrated arbiters of factual correctness.

6 From Detection to Diagnosis: Repositioning LLMs

The findings of the prior sections converge on the shared insight that LLMs alone as a detector for intrinsic hallucination lack reliability. Their capabilities are nonetheless undeniable and this section will therefore discuss two possible future directions for LLMs within the intrinsic detection paradigm.

6.1 State-of-the-Art Detection Methods

Across the last years of research, no single system dominates intrinsic hallucination detection across all domains, tasks and languages. As reviewed in Section 4, each method captures different dimensions of semantic inconsistency. In the SemEval-2024 SHROOM shared task (Mickus et al. 2024), participants were tasked to create a system for detecting hallucinations in outputs generated by a natural language generation model across multiple tasks. To establish a reference point, the organizers provided an LLM as the baseline using the Mistral-7B-Instruct-v0.2-GGUF model, prompted with a simple yes/no question about whether the hypothesis is supported by the context (Mickus et al. 2024, p. 1983). The results demonstrated that a single LLM, used as a zero-shot classifier, constitutes already as a reasonable starting point achieving an accuracy of 0.687. The best performing approaches, however, showed that combining multiple signals consistently yields the strongest results. This idea is most concretely instantiated by the winning system MetaCheckGPT (Mehta et al. 2024). In the first stage, each candidate sentence is compared against a set of stochastically generated responses following the self-consistency, see Section 4.2 . Multiple completions are sampled from models such as ChatGPT and Vectara, and each is queried whether it supports the original sentence, creating a set of independent uncertainty signals per model. In the second stage, a meta-regressor aggregates all these signals into a single hallucination score. On the SHROOM test set, this approach achieved an accuracy of 0.847 on the model-agnostic track (Mickus et al. 2024, p.1985). This gap compared to the LLM baseline further reflects the limits of relying on a single model.

Crucially, LLMs play a specific and bounded role within this architecture. They are not used as autonomous judges but as generators of consistency signals. The outputs are sampled, compared, and fed into a downstream aggregator that compensates for their individual instability. This division of labour is precisely what the preceding sections suggest is necessary. The prompt sensitivity and limitations that hinder LLMs as standalone detectors become less critical when their outputs are one signal among several, cross-checked against other models and aggregated statistically. The practical state of the art is therefore not a single model but a mixed architecture in which LLMs contribute semantic breadth while discriminative components provide the stability they lack.

6.2 Limitations of the Classification Paradigm

Despite these advances, all systems described above share a common architectural commitment: they frame hallucination detection as a classification problem, assigning a discrete label or a continuous risk score to a model output. This framing, while enabling benchmark-

ing and automated filtering, is subject to a set of principled limitations.

The most fundamental is the absence of causal explanation. A binary label flags that something is wrong but cannot communicate *what* is wrong, *where* the inconsistency occurs, or *why* the model produced it. In practice, this severely limits the utility of detection for downstream purposes. A developer correcting a generative model can not act on a single label alone while a human reviewer examining a flagged output still must perform the diagnostic work manually.

A second limitation is the lack of feedback for model improvement. Detection systems are typically applied post-hoc to finished outputs. Even when a detector achieves high accuracy, it provides no gradient signal and no structured information that could be fed back into the generation process to reduce future hallucinations. This also represents a conceptual ceiling that detection can filter outputs but cannot cure the underlying failure mode (Yanyi Liu et al. 2025, p. 1).

Finally, annotation ambiguity which, means that the binary label itself is unstable. The SHROOM shared task showed that the inter-annotator agreement across participants varied (Mickus et al. 2024, p. 1985), and the SHROOM task organisers explicitly warn that even top-performing systems can behave close to random on harder items (Mickus et al. 2024, p.1985). Compressing a complex semantic consistency judgment into a single label therefore discards these uncertainties.

6.3 Towards a Diagnosis Framework

Building upon these limitations the paper (Yanyi Liu et al. 2025) proposes to expand the hallucination detection task to a Diagnosis task. The hallucinations diagnosis model should therefore have the ability to examine the task across four dimensions.

1. **Detection** — accurately identifying whether the generated content aligns with the source context.

2. **Localization** — pinpointing the specific segments within the output that are hallucinatory.

3. **Explainability** — explaining why the identified content is inconsistent with the source.

4. **Mitigation** — proposing corrections or generating a revised version of the output that restores faithfulness to the source.

To operationalize these four diagnostic dimensions, the authors propose a two-component methodology. First, the Hallucination Diagnosis Generator (HDG), an automated data synthesis pipeline, is introduced to construct a large-scale, richly annotated training dataset from raw corpora. Second, building upon this synthesized data, the Hallucination Diagnosis Model (HDM-4B-RL), a 4-billion-parameter model, is trained using Group Relative Policy Optimization (GRPO) to guide the model toward the desired diagnostic behavior across all four dimensions.

The experimental results demonstrate the effectiveness of this approach. On the hallucination detection task, HDM-4B-RL outperforms all previous specialized detection models,

surpassing the prior state-of-the-art model Lynx-8B by 2.4% in average F1 score despite being half the size. Furthermore, the model achieves an average F1 score of 79.65, exceeding powerful closed-source models such as GPT-4.1 (75.37 F1) while operating at roughly half the inference time of the 32B-scale models (Yanyi Liu et al. 2025, p.6).

In the full hallucination diagnosis task, covering detection, localization, and mitigation, HDM-4B-RL demonstrates highly competitive performance against much larger general-purpose models. Operating as a lightweight end-to-end single-prompt model, it closely approaches the diagnostic quality of the 32B-scale Qwen3 model, while requiring only $1\times$ the runtime compared to the $4.9\times$ demanded by the pipeline-based 32B approach (Yanyi Liu et al. 2025, p.7). These results validate that a compact, specialized model, when trained on high-quality diagnostic data, can serve as an efficient and reliable alternative to large general-purpose models in hallucination analysis tasks (Yanyi Liu et al. 2025, p.7).

(Yanyi Liu et al. 2025, p.1) draws the four-dimensional framework analogous to a clinical diagnosis. A doctor must not only determine that an illness is present, but also locate it, characterise its cause, and propose a treatment, a diagnostic hallucination model must not merely flag an error but reason about it.

6.4 Where LLMs Fit: Detection, Diagnosis, and Hybrid Architectures

The shift from detection to diagnosis re-positions LLMs in an important way. In the detection paradigm, LLMs are used primarily as black-box classifiers, and their suitability is constrained by the structural weaknesses discussed in Section 5. In the diagnosis paradigm, however, LLMs are not asked to produce a single label but to engage in structured, multi-step reasoning. This is a task that aligns far better with their demonstrated strengths in chain-of-thought inference and instruction-following (Yanyi Liu et al. 2025, p.2).

The prior sections therefore implictily outline two distinct roles that LLMs can occupy with in the paradigm of detecting intrinsic hallucinations. In the first role, LLMs act as signal generators within an ensemble architecture. As demonstrated by MetaCheckGPT (Mehta et al. 2024, p.2), lies the value of LLMs not in producing a final verdict but in contributing one consistency signal among several. In the second role, LLMs act as a diagnostic reasoners. Instead of producing a binary label, LLMs are asked to localise the inconsistency, explain the error and propose a correction. The HDM-4B-RL model shows that a compact instruction-tuned LLM trained on high quality diagnostic data can outperform much larger general-purpose models on both detection and localisation. This particular role aligns more strongly with the architectural strengths of LLMs.

7 Conclusion

This paper set out to evaluate how well LLMs are suited for detecting intrinsic hallucinations, discussing that they are conditionally capable for this task, but structurally insufficient for reliable real-world deployment.

The seed paper (Gogoulou et al. 2025) established that large, instruction-tuned LLMs can perform competitively at the task of intrinsic hallucination. Nonetheless must these results

be understood as an upper bound obtained by the idealized conditions. The review of alternative approaches, like entailment-based models, self-consistency methods and embedding-based detectors each capture different signals and therefore provide a unique contribution, but none dominates across all desired tasks and languages. The practical state of the art approaches, demonstrated by the SHROOM shared task (Mickus et al. 2024, p. 1986) highlight that not a single model but a hybrid architecture in which LLMs contribute semantic signals as one component among others, yield the best performance. On the other hand, as extensively discussed, LLMs can never serve as a reliable and standalone detector. Systematic biases rooted in training and strong environmental dependent sensitivity arising from the probabilistic nature of autoregressive generation sets a limit. A system that itself hallucinates can not be a fully reliable detector. Taken together, these findings could motivate the shift from detection to diagnosis. Detection establishes that an error exists while diagnosis asks what the error is, where it occurs, and how it can be corrected. This expansion allows LLMs to no longer produce a binary verdict but perform the kind of structured, multistep semantic reasoning for which their architecture is genuinely fit for. The HDM-4B-RL model (Yanyi Liu et al. 2025) demonstrates that this is not merely a theoretical direction but rather a compact, specialised model trained on diagnostic data that can outperform much larger general-purpose systems on both detection and localisation tasks. The broader implication is that the question of whether LLMs are suited for hallucination detection does not have a single answer. Under controlled conditions, as part of a hybrid architecture, or in the role of a diagnostic reasoner, they are a powerful tool. As uncalibrated, standalone arbiters of faithfulness, they are not. Future progress in this area will depend less on finding a single better detector and more on designing systems that assign each component the role for which it is most reliable.

8 Limitations

A central challenge in research on intrinsic hallucinations in LLMs lies in the limitations of currently available datasets and evaluation methods (Gu et al. 2025, p.49). Many studies rely on relatively small or task-specific datasets, such as NLI-style benchmarks or datasets designed for particular tasks like summarization or machine translation (Gogoulou et al. 2025). Most of the datasets are often manually created and carefully altered for a specific evaluation scenario, which makes them well suited for controlled experiments but limits their ability to generalize to other tasks or domains. Conclusions drawn from such datasets are frequently highly task-dependent and may not reflect the broader phenomenon of intrinsic hallucinations in LLM outputs. Another limitation is the lack of a consistent and widely accepted taxonomy of hallucination types. As outlined, different studies adopt slightly different definitions and categorizations of hallucinations, which makes it difficult to compare results across benchmarks or to develop standardized evaluation frameworks. Furthermore, defining hallucination itself remains a challenge, as the classification of a statement often depends on what is considered the source and how the underlying context is interpreted. As mentioned in the shared SHROOM task (Mickus et al. 2024, p.1987) even humans had strong disagreement about what to flag as hallucination and that the problem in general would better

be classified as a gradient problem rather than a binary. Additionally, the majority of evaluated models and benchmarks are centered around high-resource languages such as English. Low-resource languages are barely covered. The extent to which the findings generalize to languages with less pre-training data remains largely unexplored, which indirectly restricts the conclusions drawn. Finally, there is still no universally accepted metric for reliably measuring hallucinations. Many commonly used evaluation metrics show inconsistent alignment with human judgments across datasets, models, and generation settings. This inconsistency makes it difficult to establish robust benchmarks and complicates the comparison of different hallucination detection approaches. In comparison to extrinsic hallucination, intrinsic hallucination receives comparatively little attention although the impact is undeniable. It therefore needs more extensive research.

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